

4000-byte datagram

$$MTU = 1500$$

$$\text{Actual MTU} = 1500 - 20 \\ = 1480$$

$$\text{Actual data} = 4000 - 20 \\ = 3980$$

$$D_1: 1480 + 20$$

$$D_2: 1480 + 20$$

$$D_3: 1020 + 20$$

- A Sender is transmitting an IP datagram with a total size of **4020 bytes** (4000 bytes of data + a 20-byte IP header) to a Receiver.
- The network path consists of four sequential links (N2, N3, N4, N1) separated by three routers (R1, R2, R3).
- The MTUs for the respective network technologies are:
 - **N2 (Ethernet):** 1500 bytes
 - **N3 (FDDI):** 1020 bytes
 - **N4 (PPPoE):** 1492 bytes
 - **N1 (Ethernet):** 1500 bytes

$$\text{Actual datagram} = 4020 - 20 \\ = 4000$$

N2:

$$\text{Actual MTU} = 1500 - 20 \\ = 1480$$

$$D_1: 1480 + 20$$

$$D_2: 1480 + 20$$

$$D_3: 1040 + 20$$

N3:

$$\text{Actual MTU} = 1020 - 20$$

$$= 1000$$

$$D_{1.1} = 1000 + 20$$

$$D_{1.2} = 480 + 20$$

$$D_{2.1} = 1000 + 20$$

$$D_{2.2} = 480 + 20$$

$$D_{3.1} = 1000 + 20$$

$$D_{3.2} = 40 + 20$$

N4:

$$\begin{aligned} \text{Actual MTU} &= 1492 - 20 \\ &= 1472 \end{aligned}$$

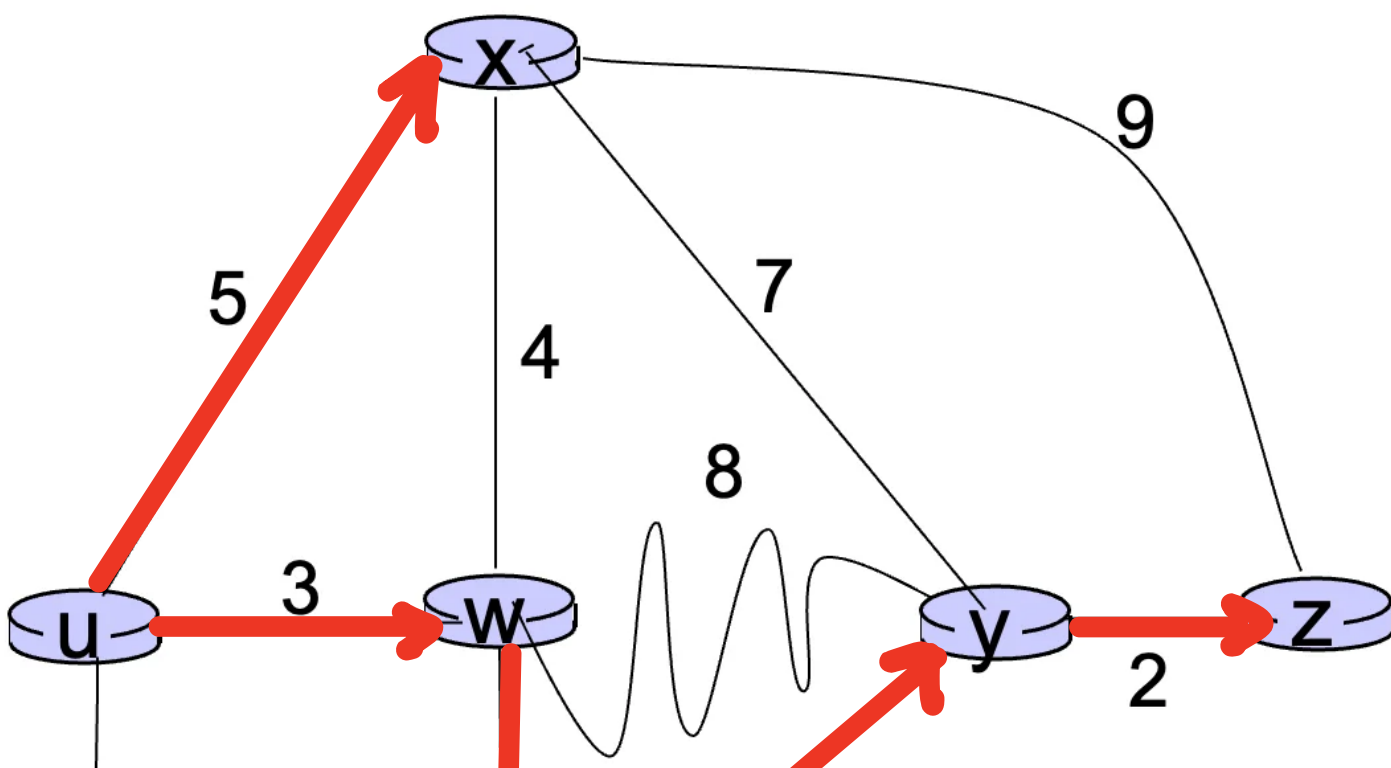
No new fragments

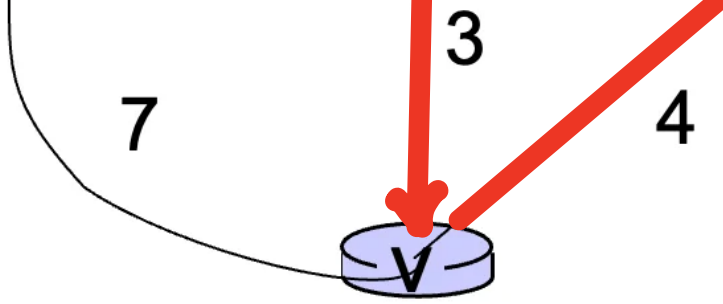
N1:

$$\begin{aligned} \text{Actual MTU} &= 1500 - 20 \\ &= 1480 \end{aligned}$$

No new fragments

Dijkstra's Link-State





N	x	w	v	y	z
u	5	<u>3</u>	7	∞	∞
uw	<u>5</u>		6	11	∞
uw x			<u>6</u>	11	14
uw xv				<u>10</u>	14
uw xv y					<u>12</u>
uw xv y z					

$$D(x) = \min(D(x), D(w) + C_{w,x}) = \min(5, 3 + 4) = 5$$

$$D(y) = \min(D(y), D(w) + C_{w,y}) = \min(\infty, 3 + 8) = 11$$

$$D(v) = \min(D(v), D(w) + C_{w,v}) = \min(7, 3 + 3) = 6$$

$$D(x) = \min(D(x), D(v) + C_{v,x}) = \min(11, 6 + 5) = 11$$

$$D(z) = \min(D(z), D(x) + C_{x,z}) = \min(\infty, 11 + 3) = 14$$

$$D(y) = \min(D(y), D(v) + C_{v,y}) = \min(11, 6 + 4) = 10$$

$$D(z) = \min(D(z), D(y) + C_{y,z}) = \min(14, 10 + 2) = 12$$

Link State Algorithm

Distance vector example:



t=1

- b receives DVs from a, c, e, computes:

DV in a:
$D_a(a)=0$
$D_a(b)=8$
$D_a(c)=\infty$
$D_a(d)=1$
$D_a(e)=\infty$
$D_a(f)=\infty$
$D_a(g)=\infty$
$D_a(h)=\infty$
$D_a(i)=\infty$

a

8

compute

1

e

DV in b:

$D_b(a)=8$	$D_b(f)=\infty$
$D_b(c)=1$	$D_b(g)=\infty$
$D_b(d)=\infty$	$D_b(h)=\infty$
$D_b(e)=1$	$D_b(i)=\infty$

DV in c:

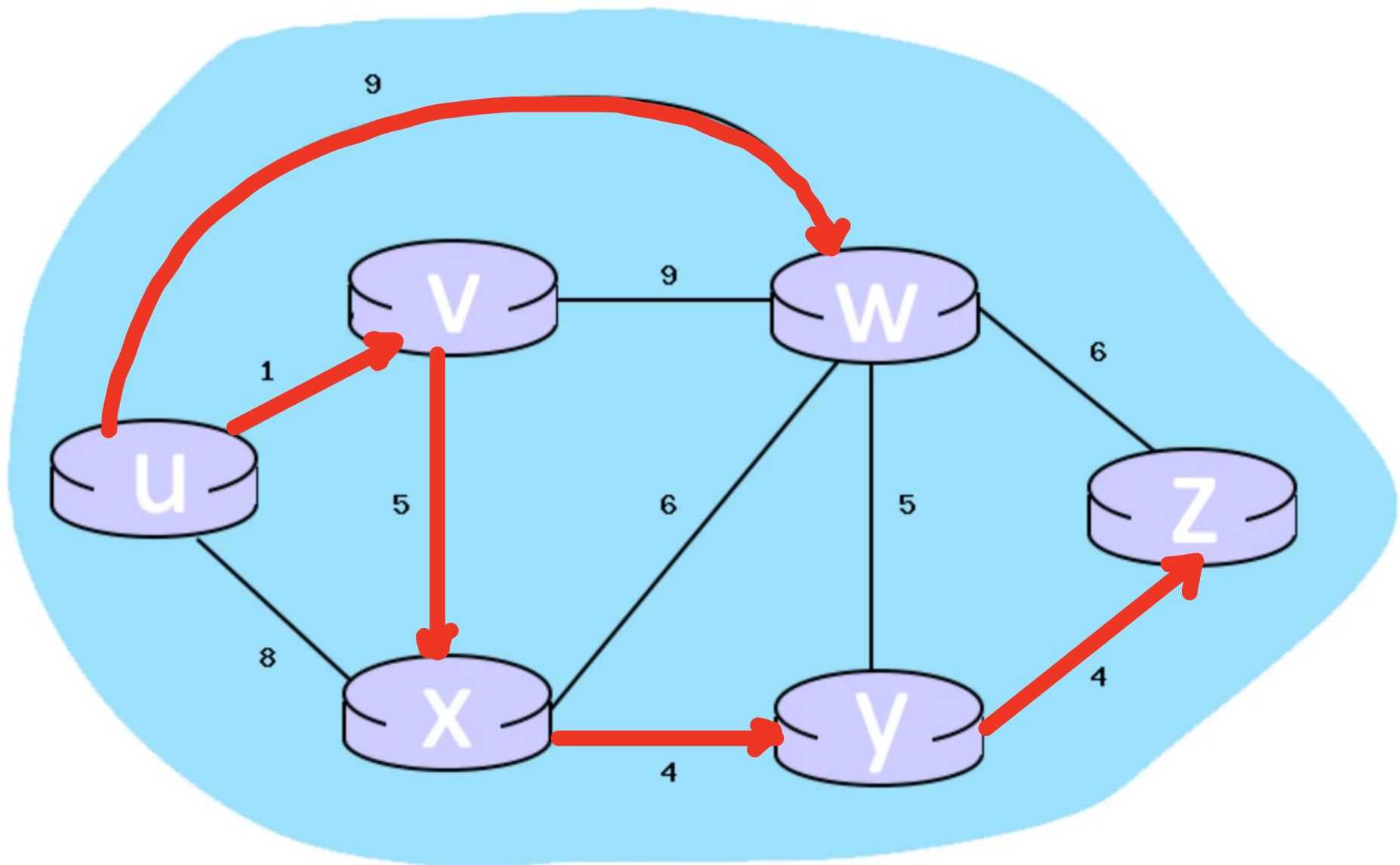
$D_c(a)=\infty$
$D_c(b)=1$
$D_c(c)=0$
$D_c(d)=\infty$
$D_c(e)=\infty$
$D_c(f)=\infty$
$D_c(g)=\infty$
$D_c(h)=\infty$
$D_c(i)=\infty$

DV in e:

$D_e(a)=\infty$
$D_e(b)=1$
$D_e(c)=\infty$
$D_e(d)=1$
$D_e(e)=0$
$D_e(f)=1$
$D_e(g)=\infty$
$D_e(h)=1$
$D_e(i)=\infty$

$$\begin{aligned}
 D_b(a) &\rightarrow 8 \\
 D_b(b) &= 0 \\
 D_b(c) &\rightarrow 1 \\
 D_b(d) &\rightarrow 2 \text{ (new)} \\
 D_b(e) &\rightarrow 1 \\
 D_b(f) &\rightarrow 2 \text{ (new)} \\
 D_b(g) &\rightarrow \infty \\
 D_b(h) &\rightarrow 2 \text{ (new)} \\
 D_b(i) &\rightarrow \infty
 \end{aligned}$$

$$\begin{aligned}
 D_b(a) &= \min \{ C_{b,c} + D_c(a), C_{b,a} + D_a(a), C_{b,e} + D_e(a) \} \\
 D_b(b) &= 0 \\
 D_b(c) &= \min \{ C_{b,c} + D_c(c), C_{b,a} + D_a(c), C_{b,e} + D_e(c) \} \\
 D_b(d) &= \min \{ C_{b,c} + D_c(d), C_{b,a} + D_a(d), C_{b,e} + D_e(d) \} \\
 D_b(e) &= \min \{ C_{b,c} + D_c(e), C_{b,a} + D_a(e), C_{b,e} + D_e(e) \} \\
 D_b(f) &= \min \{ C_{b,c} + D_c(f), C_{b,a} + D_a(f), C_{b,e} + D_e(f) \} \\
 D_b(g) &= \min \{ C_{b,c} + D_c(g), C_{b,a} + D_a(g), C_{b,e} + D_e(g) \} \\
 D_b(h) &= \min \{ C_{b,c} + D_c(h), C_{b,a} + D_a(h), C_{b,e} + D_e(h) \} \\
 D_b(i) &= \min \{ C_{b,c} + D_c(i), C_{b,a} + D_a(i), C_{b,e} + D_e(i) \}
 \end{aligned}$$



N'	$D(x)$	$D(v)$	$D(w)$	$D(y)$	$D(z)$
u	8	<u>1</u>	9	∞	∞
uv	<u>6</u>		9	∞	∞
uvx			<u>9</u>	10	∞
$uvxw$				<u>10</u>	15
$uvxwy$					14
$uvxwyz$					

$$D(w) = \min(D(w), D(v) + C_{v,w}) = (9, 1 + 9) = 9$$

$$D(x) = \min(D(x), D(v) + C_{v,x}) = (8, 1 + 5) = 6^{\text{new}}$$

$$D(w) = \min(D(w), D(x) + C_{x,w}) = (9, 6 + 6) = 9$$

$$D(x) = \min(D(x), D(y) + C_{x,y}) = (\infty, 6 + 4) = 10^{\text{new}}$$

$$D(z) = \min(D(z), D(w) + C_{w,z}) = (\infty, 9 + 6) = 15$$

$$D(y) = \min(D(y), D(w) + C_{w,y}) = (10, 9 + 5) = 10$$

$$D(z) = \min(D(z), D(y) + C_{y,z}) = (15, 10 + 4) = 14$$